```
clc;
clear;
close all;
```

INPUTS

```
b = [0.0048 0.0193 0.0289 0.0193 0.0048];
a = [1 -2.3695 2.3140 -1.0547 0.1874];

A = 0.9;
f = 50;
Fs = 1000;
N = 200;
B = 8;
SF = 0.5;

n = 0:N-1;
x = A*sin(2*pi*f*n/Fs);
S = 2^(B-1); % Q1.7
```

IDEAL AND QUANTIZED RESPONSE

```
y = filter(b,a,x);
q = @(z) round(z*S)/S;
ys = q(y);
```

OVERFLOW

```
ov = abs(y)>1;
No_Overflow = sum(ov);
Peak_Unscaled = max(abs(y));
```

SATURATION

```
ysat = q(min(max(y,-1),1));
```

WRAP-AROUND

```
ywrap = (mod(round(y*S)+S,2*S)-S)/S;
```

SIGNAL SCALING

```
xs = SF*x;
yscale = q(filter(b,a,xs));
Peak_Scaled = max(abs(yscale));
No_Overflow_Scaled = sum(abs(filter(b,a,xs))>1);
```

ERRORS

```
RMSE_Sat = sqrt(mean((y-ysat).^2));
RMSE_Wrap = sqrt(mean((y-ywrap).^2));
RMSE_Scale = sqrt(mean((SF*y-yscale).^2));
```

OUTPUT

```
fprintf('\n===== RESULTS =====\n');
fprintf('Q1.7 Range : [-1, 0.9921875]\n');
fprintf('Unscaled Peak : %.6f\n',Peak_Unscaled);
fprintf('Overflow Count : %d\n',No_Overflow);
fprintf('Saturation Peak : %.6f\n',max(abs(ysat)));
fprintf('Wrap-around Peak : %.6f\n',max(abs(ywrap)));
fprintf('Scaling Factor : %.2f\n',SF);
fprintf('Scaled Peak : %.6f\n',Peak_Scaled);
fprintf('Scaled Overflow Count : %d\n',No_Overflow_Scaled);
fprintf('Saturation RMSE : %.6e\n',RMSE_Sat);
fprintf('Wrap-around RMSE : %.6e\n',RMSE_Wrap);
fprintf('Scaled Output RMSE : %.6e\n',RMSE_Scale);
```

```
===== RESULTS =====
Q1.7 Range : [-1, 0.9921875]
Unscaled Peak : 0.901600
Overflow Count : 0
Saturation Peak : 0.898438
Wrap-around Peak : 0.898438
Scaling Factor : 0.50
Scaled Peak : 0.453125
Scaled Overflow Count : 0
Saturation RMSE : 2.389472e-03
Wrap-around RMSE : 2.389472e-03
Scaled Output RMSE : 2.219846e-03
```

TABLE

```
Method = {'Unscaled';'Saturation';'Wrap-around';'Scaled'};
Peak = [Peak_Unscaled;max(abs(ysat));max(abs(ywrap));Peak_Scaled];
RMSE = [0;RMSE_Sat;RMSE_Wrap;RMSE_Scale];
Overflow = [No_Overflow;0;0;No_Overflow_Scaled];

disp(table(Method,Peak,RMSE,Overflow));
```

Method	Peak	RMSE	Overflow
{'Unscaled' }	0.9016	0	0
{'Saturation' }	0.89844	0.0023895	0
{'Wrap-around'}	0.89844	0.0023895	0

```
    {'Scaled'     }    0.45312    0.0022198        0
```

PLOTS

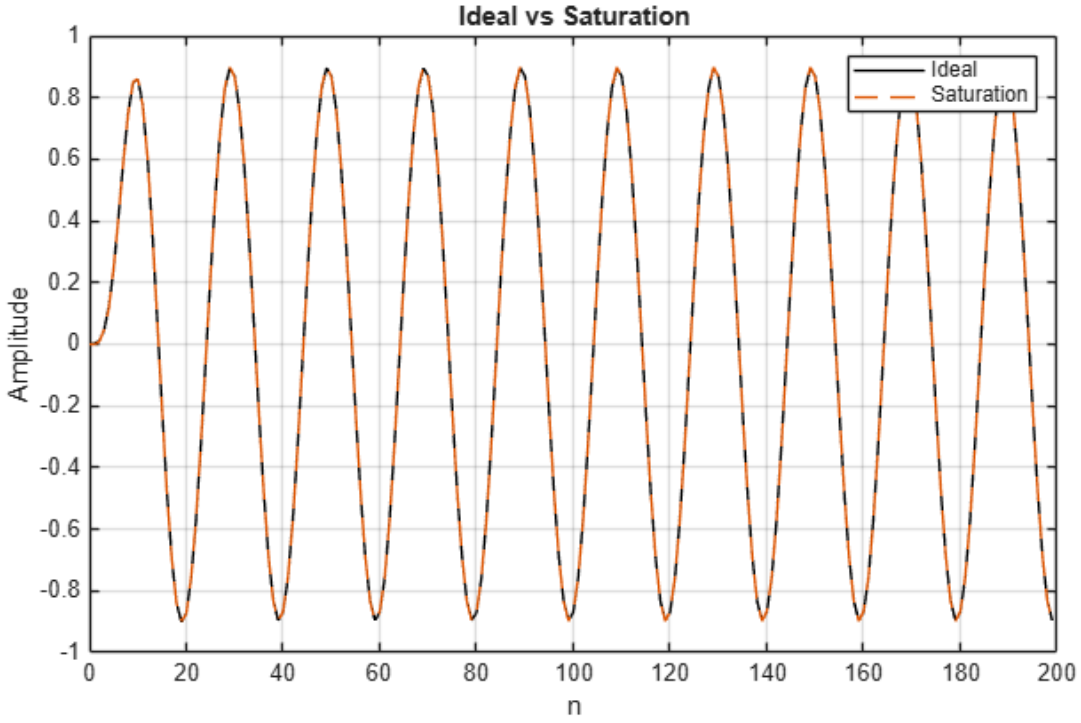
```
figure;
plot(n,y,'k',n,ysat,'--');
grid on;
title('Ideal vs Saturation');
xlabel('n');
ylabel('Amplitude');
legend('Ideal','Saturation');

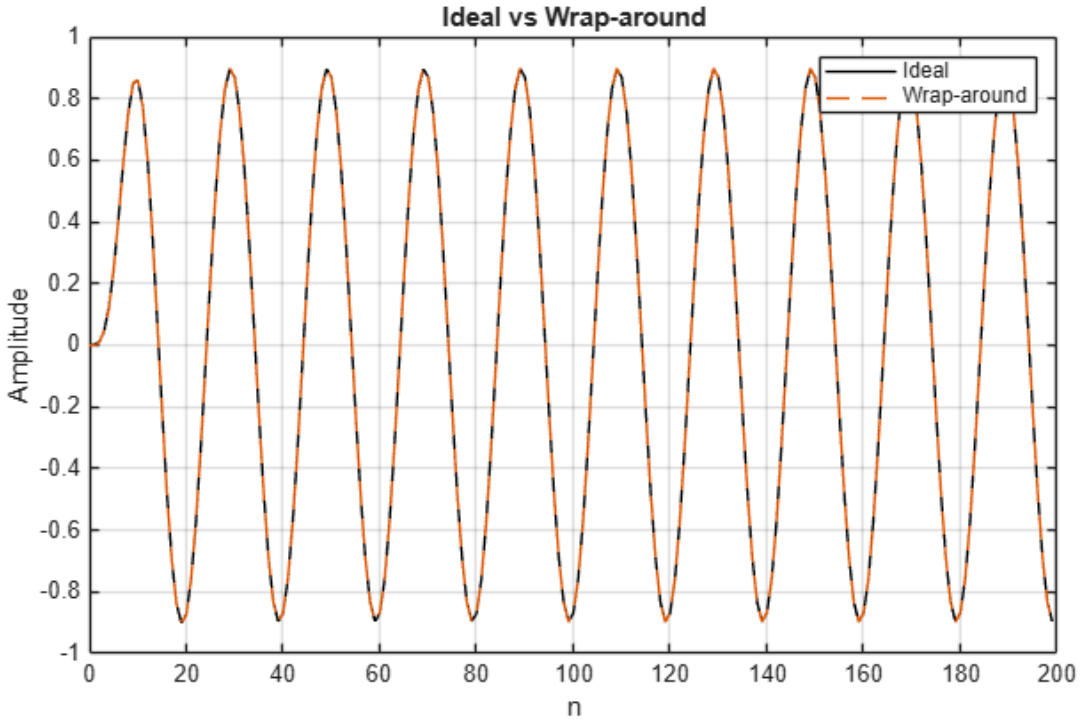
figure;
plot(n,y,'k',n,ywrap,'--');
grid on;
title('Ideal vs Wrap-around');
xlabel('n');
ylabel('Amplitude');
legend('Ideal','Wrap-around');

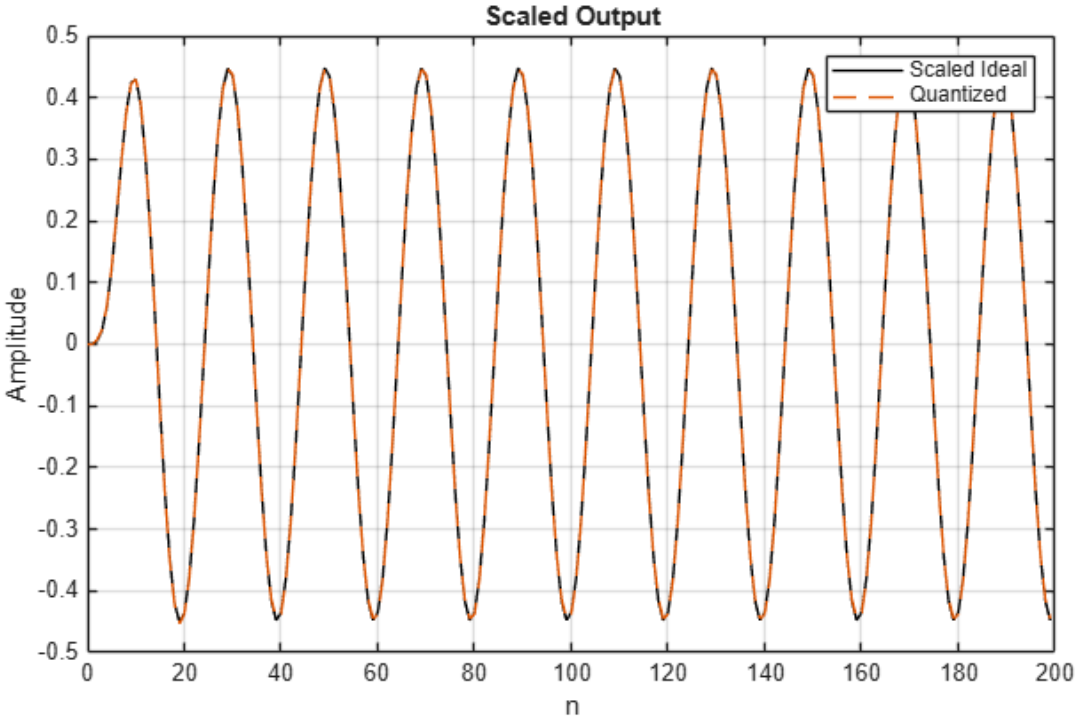
figure;
plot(n,SF*y,'k',n,yscale,'--');
grid on;
title('Scaled Output');
xlabel('n');
ylabel('Amplitude');
legend('Scaled Ideal','Quantized');

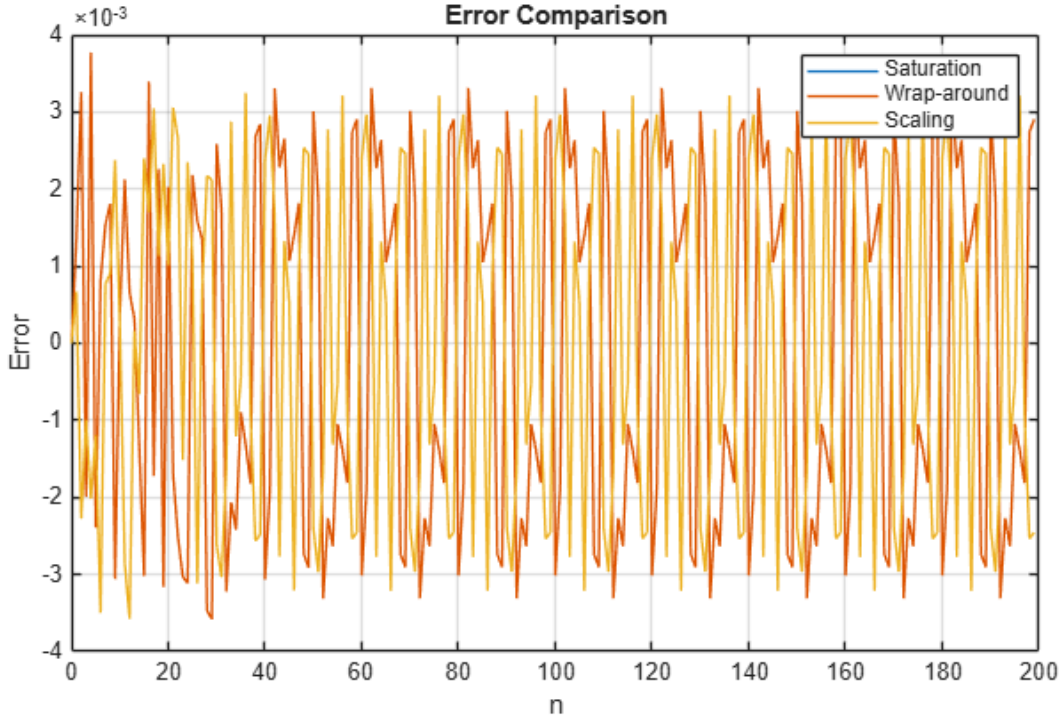
figure;
plot(n,y-ysat);
hold on;
plot(n,y-ywrap);
plot(n,SF*y-yscale);
grid on;
title('Error Comparison');
xlabel('n');
ylabel('Error');
legend('Saturation','Wrap-around','Scaling');

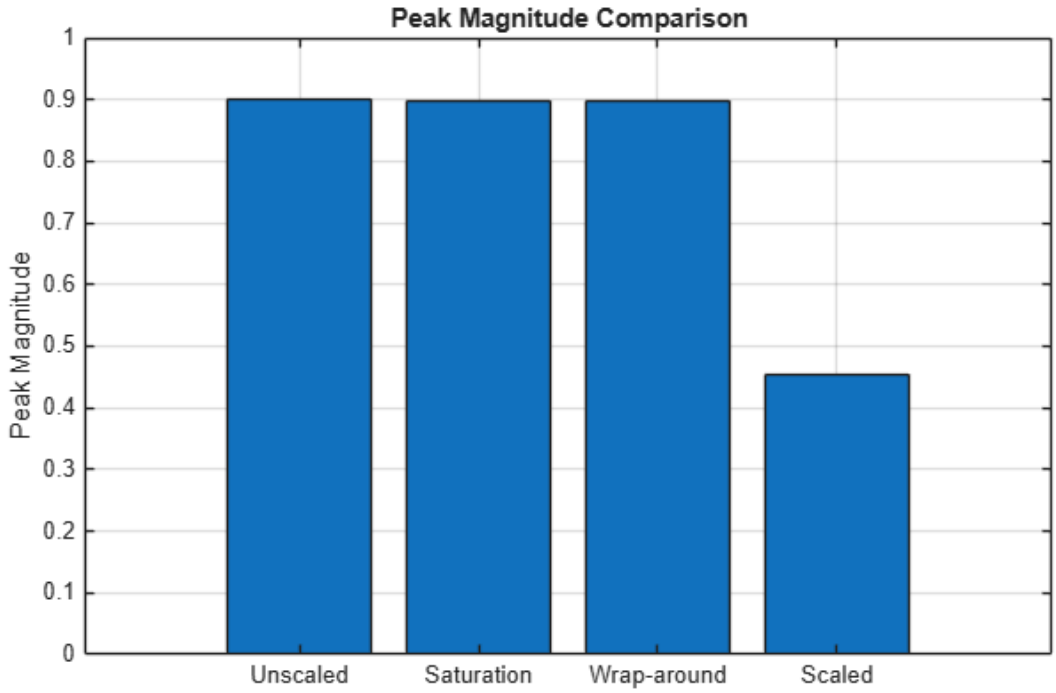
figure;
bar(Peak);
grid on;
set(gca,'XTickLabel',Method);
ylabel('Peak Magnitude');
title('Peak Magnitude Comparison');
```


Ideal vs Saturation
Ideal
Saturation
Amplitude
n
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
0
20
40
60
80
100
120
140
160
180
200


Ideal vs Wrap-around
Ideal
Wrap-around
Amplitude
n
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
0
20
40
60
80
100
120
140
160
180
200


Scaled Output
Scaled Ideal
Quantized
Amplitude
n
0.5
0.4
0.3
0.2
0.1
0
-0.1
-0.2
-0.3
-0.4
-0.5
0
20
40
60
80
100
120
140
160
180
200


Error Comparison
×10-3
Saturation
Wrap-around
Scaling
Error
n
4
3
2
1
0
-1
-2
-3
-4
0
20
40
60
80
100
120
140
160
180
200



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